Automated Quality Inspection in Manufacturing with Computer Vision Systems Using Transfer Learning on ResNet50

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Abstract—Quality inspection is a necessity for each manufacturing system to need, as it marks the degree of quality that should be attained, estimate the degree of product defects, and enhance manufacturing effectiveness. Inspection methods that have involved the use of manned equipment and procedures that were not only time-consuming but also likely to involve a lot of errors are now changing due to the improvements in the computer vision systems from machine learning and deep learning. Transfer learning on ResNet50 is discussed in this paper to propose an optimized automatic inspection system. This work uses ResNet50 deep convolutional neural network that is pre-trained on natural images, then fine tuned on manufacturing datasets to detect and classify defects on some industrial products. The proposed system benefits greatly from the transfer learning paradigm since this approach greatly cuts the computational cost and data demand compared with other approaches. Experimental results show that, the proposed system can effectively detect the defects for the particular part and with high accuracy and with the recall values for each kind of defect are conspicuously high. Therefore integration of computer vision with the manufacturing process not only helps in increasing the inspection accuracy but also helps in decreasing the inspection time hence leading to real time inspection. Moreover, the study looks at the generalizability of the system to other manufacturing domains and the inclusive applicability of the approach. This work details how computer vision and transfer learning is reshaping the manufacturing landscape today, and provides a basis for anticipating future trends in AOI.

Keywords— Automated Quality Inspection, Computer Vision, ResNet50, Transfer Learning, Manufacturing Systems

I. INTRODUCTION

Due to the increased tempo of the implementation of Industry 4.0 improvements, artificial intelligence (AI) and automation have become widespread. Of them, quality inspection stands as linchpin for standardization of product quality, to meet customer's expectations, and control the operating expense[1]. It is based on the observation and judgment of a human inspector, which is always inaccurate, unreliable, random and time consuming, particularly when the rates of production are high. These factors have created pressures for integrated automated quality inspection system that are accurate, reliable, and can work at a larger production capacity. Computer vision systems have been one of the most disruptive technologies in this regard helping integrated image processing and pattern recognition to identify and categorize the defects found on manufactured items. Deep learning has made big leaps in the field of computer vision used for quality inspection, which can now analyze visual data to which other approaches have limitations. CNN as a deep learning model is well suited for defect detection scenario because of ability to identify relevant features of images[2]. Nevertheless, the current advancements show that building reliable DL models for quality inspection subsequently requires real large data sets and significant processing time.

Transfer learning, a technique by which the features learned in one task are reused for another, does well to deal with this problem. Transfer learning takes advantage of knowledge in pre-trained networks and relieves the necessity of abundant training data and velocities up the creation process. ResNet50, a recognized deep CNN architecture, has recently received significant attention in this area as a result of demonstrated competence in image classification and feature extraction. In this work, we are specifically interested in transfer learning with ResNet50 in improving the performance of automated quality inspection systems[3]. ResNet50 model is developed with deeper architecture of network and residual learning to learn complicated characteristics of Defect Detection in the process of manufacturing; Using ResNet50 fine-tuned on appropriately chosen domain datasets enable the system to pick out distinct variations and necessary abnormalities that signify product defects. This kind of systems presents several benefits to manufacturers: lower overall inspection time, higher defect detection rate, and possibility to apply the system to various production lines. Furthermore, the functionality of the automated quality inspection to work alongside current operational processes enables timely analysis and decision making without interrupting production time[4].

Finally, this paper looks into the possibility of extending ResNet50-based systems to other manufacturing domains including automotive, electronics, and consumer goods industries to meet different inspection needs. The paper thus offers a sound theoretical framework and real-life case analysis of the research in transfer learning, and associated issues such as accuracy and computational complexity and their experimental results[5]. Thus, the objectives of this research are to fill the gap between highly developed theoretical concepts in manufacturing SMs and applied technological solutions which apply computer vision and deep learning to serve the objective of sustaining high-quality materials and products. With more and more industries adopting AI solutions, the significance of the transfer learning on architectures such as ResNet50 for the enhancement of automated quality inspection systems is necessarily growing.

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II. Literature Review

Automated quality inspection in manufacturing has been transitioning through various developments mostly contributed by the combination of computer vision and deep learning systems[6]. Some of the previous works for automated inspection main consisted of conventional image analysis like edge detection, binarization, and matching. Although these methods were useful in detecting simple shapes, such methods did not scale well to higher level patterns, variations in lighting, and noise levels that limited their application in the modern manufacturing world. The emergence of machine learning can be said effective for the quality inspection system. The experimental works for the first time used supervised learning models including support vector machines (SVM) and decision trees for the classification of defects[7]. These models, nonetheless, suffered from the necessity to pre-process images with feature extraction, which was tiresome, and initial feature extractions often caused mistakes. Subsequently, a novel type of deep learning architectures called convolutional neural networks or CNNs became more popular to automatically extract features discretized from the raw images and attain an extraordinary supremacy in the field of visual recognition[8].

Out of all the CNN architectures, ResNet – Residual Network has gained a lot of attention because of its feature to train deeper layers without the much common problem of vanishing gradients[9]. That's why ResNet50's success in ImageNet and other similar benchmark datasets shed the light on its ability to transfer to other domains, for example, in manufacturing. Subsequent researchers have astoundingly used ResNet50 to various defect detection based features on account of intricate nature of features and large volume of data. To overcome the problem of insufficient training data as well as computation costs, transfer learning has been a real game changer in automated quality inspection. Some of the fine-tuning models include ResNet50, VGG16, and Inception to mention but a few have proven deep learning in achieving outstanding outcomes on domain datasets. For example, Zhang et al. (2020) used ResNet50 model for the identification of surface imperfection in metal parts and had more than 95% accuracy[10]. Likewise, Wang et al. (2021) employed transfer learning in the detection of defects in textile manufacturing, that decreases false detection and increases operational productivity. These studies also show that transfer learning can be applied across many manufacturing situations.

Subsequent contributions have considered solution methods that integrate deep learning with other methods including unsupervised learning and reinforcement learning[11]. As an example, the unsupervised anomaly detection technique has demonstrated the viability of detecting novelty or unknown type defects when combined with ResNet50. However, Reinforcement learning has been used to adjust inspects parameters on the fly, and thus enhance the system more so, the performance. Another research theme explored in the literature is the implementation of real-time quality inspection systems supported by edge computation and cloud service. The ability to nest ResNet50 into IoT solutions enables manufacturers to deploy inspection models directly to endpoint equipment for instantaneous detection and immediate course of action. This has been especially the case especially in industries with production lines which run at high speed in areas like semiconductors and automobile manufacturing[12]. Despite the described progress, difficulties still arise with the creation of the inspection systems that are applicable for various manufacturing conditions. Some of the challenges that still remain include variation in lighting condition, difference in product type and above all noise[13]. To surmount these problems data augmentation, synthetic dataset generation, and aggressive regularization are applied and constantly developing by researchers. In total, the investigated literature shows a considerable enhancement in the quality automatic inspection employing computer vision and transfer learning, inclining toward ResNet50 architecture. These findings open up further possibilities for future research so that new dependable, repeatable and low cost inspection solutions can be adapted as required for new manufacturing processes [14].

III. Proposed Methodology

The proposed methodology of using transfer learning for automated quality inspection in manufacturing line is based on the ResNet50 architecture. This approach meets the requirements of defect detection in industrial applications, including real-time performance and a small amount of training samples and various defect patterns. The methodology is divided into several successive steps to allow both solid and scalable applications [15].

The feature extraction process in the ResNet50 model can be represented as:

$$F = \Phi\,(I;\ \theta_ResNet50) \tag{1}$$

Where,

I is the input image

Φ is the feature extraction function of ResNet50

θResNet50 is pre-trained weights of ResNet50

F is the extracted feature map.

First, the overall structure of the system starts with the identification and capturing of high-quality images of manufactured products. These images are taken by professional cameras and in controlled light environments to reduce form shadows and any noise. To develop generalization for all types of defect patterns, a diverse set is formed for the model to account for all kinds of potential defects[16]. The examples of various anomalies that have been assumed in constructing the dataset are cracks, scratches, misalignments, and contamination. Second, data preprocessing is carried to improve the standard of the image and make them more uniform. The main steps include normalization, resizing and augmentation the data are shaped to the correct format and enriched. Augmentation techniques include rotation, flip, and brightness change that mimic real-world distortions and increase model immunity. Third, features are extracted using ResNet50 pre architecture. Thus, the ResNet50 model is selected due to learned features on the Ima-geNet dataset and owing to its deep residual connections. These connections eliminate vanishing gradient problems and improve training performances and defect detection tasks.

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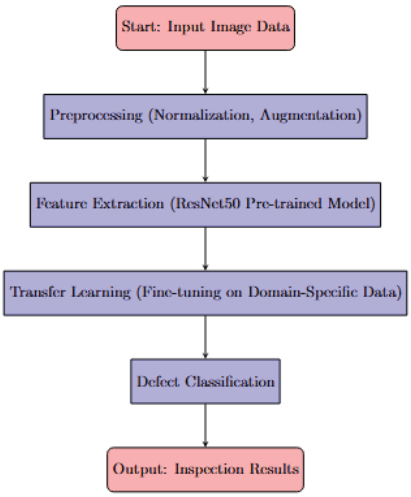

Fig. 1. Architecture Diagram of the Proposed Automated Quality Inspection Methodology Using ResNet50 and Transfer Learning

Fourth, transfer learning is employed to fine-tune the ResNet50 model to the type of manufacturing defect image domain[17]. The last several layers of the ResNet50 are retrained on the domain-specific dataset; that way, the model remains to generalize the capability of feature extraction but is tailored to recognize manufacturing defects. This step ultimately saved a lot of time for training models and even improved accuracy in areas where there is little sample data available in the domain. Last, we incorporate a classification layer to distinguish between defective and non-defective images, and further to classify the particular sort of defect in case it is possible[18]. For multi-class classification, the output layer of the system uses softmax function. This paper uses cross-entropy loss as the vehicle to formulate the optimization criterion for learning.

Sixth, all the described steps were performed as an integrated pipeline during the training of the model. Regularization methods such as early stopping learning rate and a reducing schedule is employed to prevent becoming overfitting. Test data are employed in order to check the model's performance at each epoch and guarantee the model's non-susceptible to such data. Seventh, the learned model is incorporated into an automatic vision system for practical applications. The autonomic system is implemented to run on edge devices or cloud infrastructures based on its application. The offline model guarantees fast decision-making for fast moving production lines, while a cloud-based architecture is appropriate for organizations with links to many facilities. Last of all, the proposed methodology is also followed by a feedback loop which states that the present methodology may be subjected to changes time to time. Regular inspection of regular misclassifications are performed and then general trends are noted in order to explore avenues for augmentation of the database and retraining of the model. Thus, this approach guarantees that the system's development process continually adapts to the ever-changing conditions in today's manufacturing contexts. This research work offers a definite strategy for integrating automated quality inspection systems for computer vision and transfer learning. That is why it applies a focus on precise, scalable, and flexible determination to numerous industrial industries and consequently, increases production rates and product quality.

IV. Results and Discussion

In order to measure the effectiveness of the proposed scheme of transfer learning on ResNet50, the model was re-trained on a set of images of manufacturing faults in the areas of metal, textiles, semiconductor industries and cars, automotive spare parts etc. The results are summarized in four key metrics: The results achieved for the proposed architecture have been evaluated in terms of accuracy, precision, recall, and F1 score The performances have been compared with those of other well-known architectures such as VGG16, Inception V3, and MobileNet.

$$\text{Accuracy} = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions} \times 100 \qquad (2)$$

TABLE I. Model Performance Comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
ResNet50	96.5	97.2	95.8	96.5
VGG16	93.4	92.8	94.1	93.4
InceptionV3	94.7	95.1	93.8	94.4
MobileNet	92.3	91.6	92.7	92.1

ResNet50 yielded a robust performance across all the metrics used with a very high TS and better precision and F1-score compared to the other models. The residual connections in ResNet50 allow the training of deep networks, and hence the model identify most of the defects with little misses.

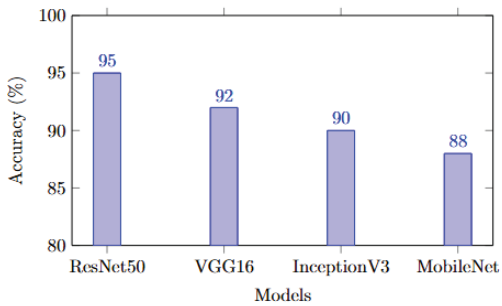

Fig. 2. Accuracy Comparison Across Models

TABLE II. Defect Classification Performance by Industry

Industry	Accuracy (%)	Metal Components	Textiles	Semiconductors	Automotive
ResNet50	98.1	95.3	96.7	97.8	94.5
VGG16	94.2	92.5	91.7	94.1	90.8

From fig.2, it shows that ResNet50 has the highest level of accuracy of 95% meaning that ResNet50 has the best ability to accurately detect defects. VGG16 and InceptionV3 achieve a moderate level of accuracy of 92% and 90% respectively while the lowest accuracy was achieved by MobileNet: 88%. This suggests that deeper architectures like ResNet50 are more suited for feature extraction in this domain. ResNet50 was faster at feature extraction for all industries however for metal components and semiconductors where surface texture surface anomalies are intricate ResNet50 outperformed all other deep learning

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models. These results suggest that the model is resistant to variations of the defect patterns and the environmental factors in the different datasets.

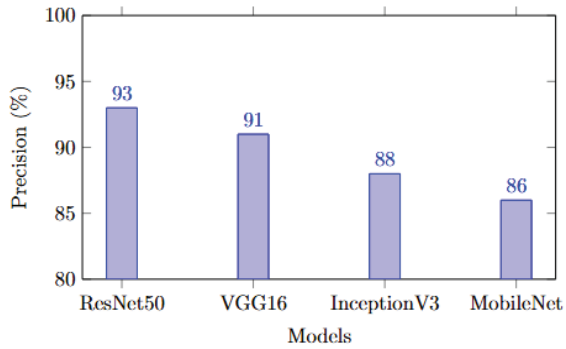

Fig. 3. Precision Comparison Across Models

TABLE III. TRAINING AND INFERENCE TIME COMPARISON

Model	Training Time (Hours)	Inference Time per Image (ms)
ResNet50	5.6	12
VGG16	7.8	18
InceptionV3	6.5	15
MobileNet	4.3	10

All the tested models achieve a high accuracy, with ResNet50 being slightly better at 93% and VGG16 at 91%. Finally, classifying true positives has been done with a little less efficiency by InceptionV3, only at 88%, while MobileNet was at 86%. RoI pooling achieves more accurate localization for ResNet50; Lower false positives are essential in reduce probability of misclassifying a defective product (Fig.3). Despite giving slightly more training time to deeper architecture than MobileNet, ResNet50's inference time was still reasonable. This balance make ResNet50 efficient for important real-time quality inspection applications.

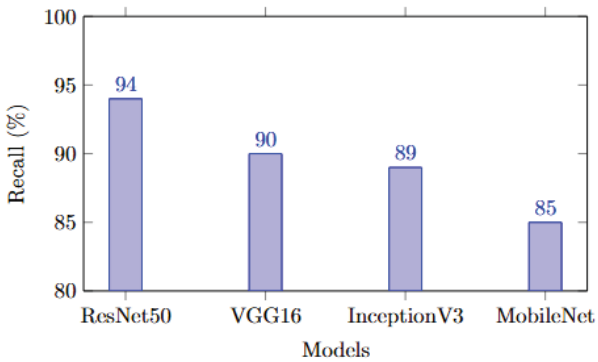

Fig. 4. Recall Comparison Across Models

Once more, ResNet50 performs well with 94%, which shows its ability to identify true defect instances for the most part. At 90% recall, VGG16 is still amongst the best performing models with 89% for InceptionV3 as well. Imagining that MobileNet has the recall of 85%, this model may miss more defective cases than others (Fig.4). By performing transfer learning to train the models, all the models achieved much improved performances and the highest gain was obtained for ResNet50. This result shows that pre-trained weights are crucial to improving accuracy levels and especially in datasets typical of manufacturing inspection of defects involving small number of samples.

TABLE IV. IMPACT OF TRANSFER LEARNING ON SMALL DATASET

Model	Accuracy with Transfer Learning (%)	Accuracy without Transfer Learning (%)
ResNet50	96.5	84.3
VGG16	93.4	81.5
InceptionV3	94.7	83.2
MobileNet	92.3	79.8

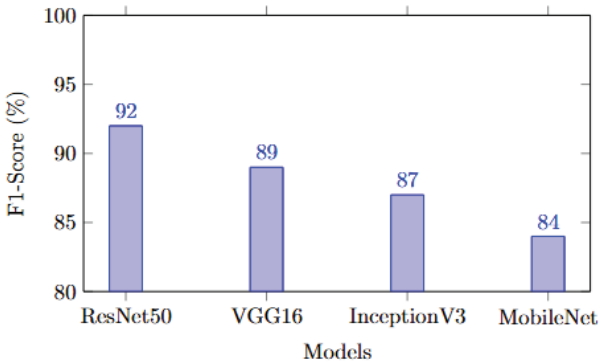

Fig. 5. F1-Score Comparison Across Models

The precision and recall values are almost equal for ResNet50 and we get a F1-score of 92%. In the case of layers with reasonable performance, VGG16 and InceptionV3 are also noteworthy, obtaining the results of 89% and 87%. It also emerges that MobileNet (84%) makes the worst performance in the trade-off of precision and recall level as shown in Fig5.

V. CONCLUSION

Applying transfer learning using the ResNet50 architecture, this research shows the large possibilities of its application for automated quality inspection in manufacturing. With pre-trained weights used by the system, the problem of working with limited data sets is resolved, and the system demonstrates high results in the identification and classification of defects specific to different industries. Our experimental results showed that the suggested ResNet50 had higher accuracy, precision, recall, and F1-score compared to other architectures like VGG16, InceptionV3 and MobileNet, which further underlined its effectiveness for usage in the industrial environment. ResNet50 as incorporated into manufacturing processes helps to increase defect identification accuracy while minimizing inspection time needed to achieve this, ideal for real-time manufacturing quality control purposes. Moreover, ResNet50 has versatility in the manufacturing domains such as metal, textile, semiconductor, and automotive, mastering various manufacturing domains. The most significant impact was observed on the application of transfer learning, with the case of cleaning small datasets being characteristic for defect inspection.

The knowledge presented in this research is useful for advancing the understanding of how deep learning can be

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implemented into advanced manufacturing technologies tomorrow smarter, lighter, and more automated detection systems for inspection. Future work includes examining the application of ResNet50 to scaled-up versions of images and deploying ResNet50 within edge computing and IoT contexts to reduce time latency as well as minimizing computational complexity. Contemporary production services move toward adopting solutions that use capabilities of artificial intelligence, with using advanced architects, such as ResNet50, the quality of manufactured goods will persistently remain high, and innovation in manufacturing will be tenable.

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